

Unit1 -- BEVS

SECTION A — 2 MARKS QUESTIONS

(Answer briefly, covering every topic of Unit 1)

Topic 1: Characteristics of Living Organisms

A1. Define a cell and name the scientist credited with its discovery.

Ans: A cell is the basic structural and functional unit of all living organisms. The English natural philosopher Robert Hooke is credited with discovering cells in 1665, having observed thin slices of cork under a compound microscope and named the small box-like compartments he saw "cells."

A2. List any four characteristics of living organisms.

Ans: Growth, Reproduction, Metabolism, and Homeostasis. (Other valid answers: cellular organisation, consciousness/responsiveness, adaptation, movement, definite life span.)

Topic 2: Cell Structure and Organelles

A3. State two structural differences between prokaryotic and eukaryotic cells.

Ans: (i) Prokaryotic cells lack a true, membrane-bound nucleus (DNA lies free in the nucleoid); eukaryotic cells possess a true nucleus. (ii) Prokaryotic cells lack membrane-bound organelles (mitochondria, ER, Golgi); eukaryotic cells possess them.

A4. Why is mitochondria known as the powerhouse of the cell?

Ans: Mitochondria are called the "powerhouse of the cell" because they are the principal site of ATP generation through aerobic respiration (oxidative phosphorylation), supplying the energy currency required for almost all cellular activities.

A5. Write down any two important functions of mitochondria.

Ans: (i) Generate ATP via aerobic cellular respiration. (ii) Possess their own DNA and ribosomes and play a key role in regulating programmed cell death (apoptosis).

A6. What are "suicide bags" of the cell? Why are they termed so?

Ans: Lysosomes are termed "suicide bags" of the cell. They contain hydrolytic (digestive) enzymes that, if released due to rupture of the lysosomal membrane, digest the cell's own organelles and macromolecules, causing the cell to self-destruct (autolysis).

A7. Define ribosome. Write the functions of ribosome.

Ans: A ribosome is a small, dense, non-membrane-bound cell organelle composed of ribosomal RNA (rRNA) and protein, found free in the cytosol or attached to the rough endoplasmic reticulum. Function: it is the site of protein synthesis, translating the mRNA sequence into a polypeptide chain.

A8. Write down two differences between rough endoplasmic reticulum and smooth endoplasmic reticulum.

Ans: (i) RER is studded with ribosomes on its outer surface, giving it a rough appearance; SER lacks ribosomes and appears smooth/tubular. (ii) RER's chief function is protein synthesis, folding and modification; SER's chief function is lipid/steroid synthesis, detoxification of drugs and toxins, and calcium ion storage.

A9. State the principal function of the Golgi apparatus.

Ans: The Golgi apparatus modifies, sorts and packages proteins and lipids received from the endoplasmic reticulum into membrane-bound vesicles and dispatches them to their correct destination — the plasma membrane, lysosomes, or for secretion outside the cell.

Topic 3: Comparative Cell Biology

A10. State any two differences between plant cells and animal cells.

Ans: (i) Plant cells possess a rigid cellulose cell wall and chloroplasts; animal cells possess neither. (ii) Plant cells have one large central vacuole; animal cells have small or no vacuoles.

A11. Write down any two differences between animal cells and bacterial cells.

Ans: (i) Animal cells are eukaryotic with a true, membrane-bound nucleus; bacterial cells are prokaryotic with a nucleoid (no nuclear membrane). (ii) Animal cells possess membrane-bound organelles (mitochondria, ER, Golgi); bacterial cells lack them.

Topic 4: Carbohydrates**A12. Mention two examples of disaccharides with their constituent monosaccharides.**

Ans: Sucrose = Glucose + Fructose (joined by an α -1,2 glycosidic bond); Lactose = Glucose + Galactose (joined by a β -1,4 glycosidic bond). [Maltose = Glucose + Glucose, α -1,4 bond, is a third valid example.]

A13. What is a glycosidic bond?

Ans: A glycosidic bond is a covalent bond formed by a condensation reaction (loss of a water molecule) between the anomeric carbon of one monosaccharide and a hydroxyl group of another sugar. It links monosaccharides together to form disaccharides, oligosaccharides and polysaccharides.

A14. Write down any two functions of carbohydrates.

Ans: (i) Serve as the primary source of energy — glucose oxidation via glycolysis and the citric acid cycle yields ATP. (ii) Provide structural support (cellulose in plant cell walls, chitin in fungal walls) and mediate cell-cell recognition via glycoproteins/glycolipids.

Topic 5: Lipids**A15. Differentiate between saturated and unsaturated lipids.**

Ans: Saturated fatty acids contain no carbon-carbon double bonds, pack tightly, and are solid at room temperature (e.g., palmitic acid, stearic acid). Unsaturated fatty acids contain one or more C=C double bonds, cannot pack tightly, and are liquid at room temperature (e.g., oleic acid, linoleic acid).

A16. Write down any two functions of lipids.

Ans: (i) Energy storage — triacylglycerols yield about 9 kcal/g, more than twice the energy density of carbohydrates or proteins. (ii) Structural role — phospholipids and cholesterol form the lipid bilayer of all biological membranes.

A17. Draw the bilayer structure of a lipid.

Ans: A phospholipid bilayer is formed of two layers of phospholipid molecules arranged tail-to-tail. The hydrophilic (polar) phosphate heads face outward toward the aqueous medium on both sides, while the hydrophobic (non-polar) fatty-acid tails face inward, shielded from water. This amphipathic self-assembly forms the structural basis of every biological membrane.

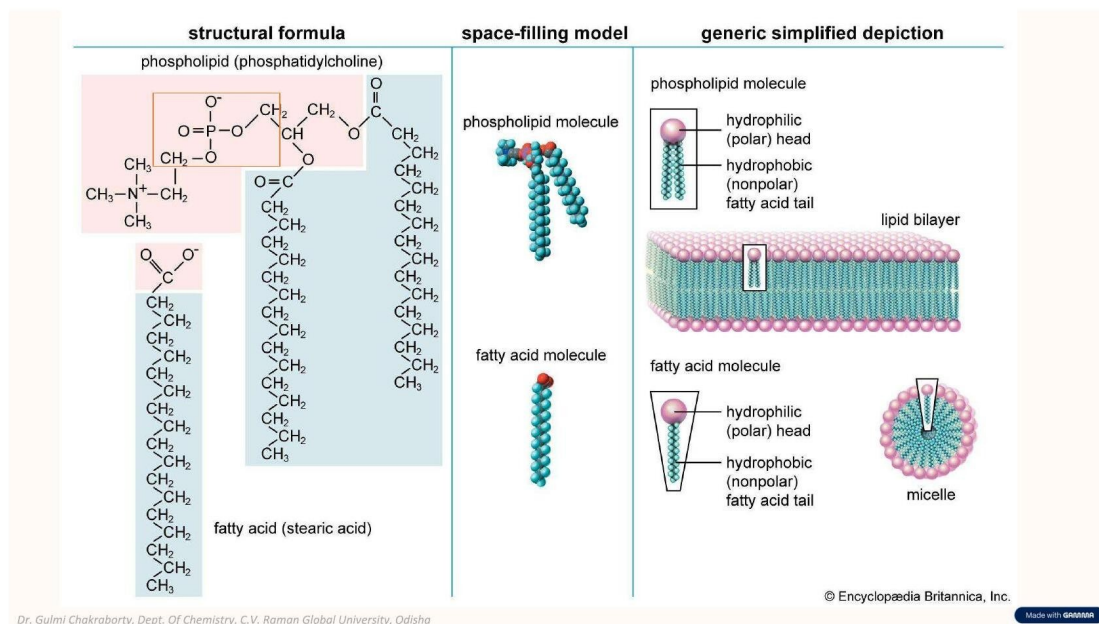


Fig. A17 — Phospholipid/fatty acid molecule and the resulting lipid bilayer / micelle

Topic 6: Proteins and Amino Acids

A18. What are glycoprotein and lipoprotein?

Ans: Glycoprotein: a conjugated protein in which the protein (apoprotein) is covalently linked to one or more carbohydrate (oligosaccharide) chains; functions in cell-cell recognition, adhesion and immune response (e.g., immunoglobulins, mucins). Lipoprotein: a complex in which protein (apolipoprotein) is non-covalently associated with lipids, enabling transport of hydrophobic lipids through aqueous blood (e.g., HDL, LDL, chylomicrons).

A19. Give an example each of an acidic, basic, neutral, sulphur-containing, and aromatic amino acid. Mention their three-letter and single-letter codes.

Ans: Acidic: Glutamic acid (Glu, E). Basic: Lysine (Lys, K). Neutral/polar-uncharged: Serine (Ser, S). Sulphur-containing: Cysteine (Cys, C). Aromatic: Phenylalanine (Phe, F). Structures are shown below.

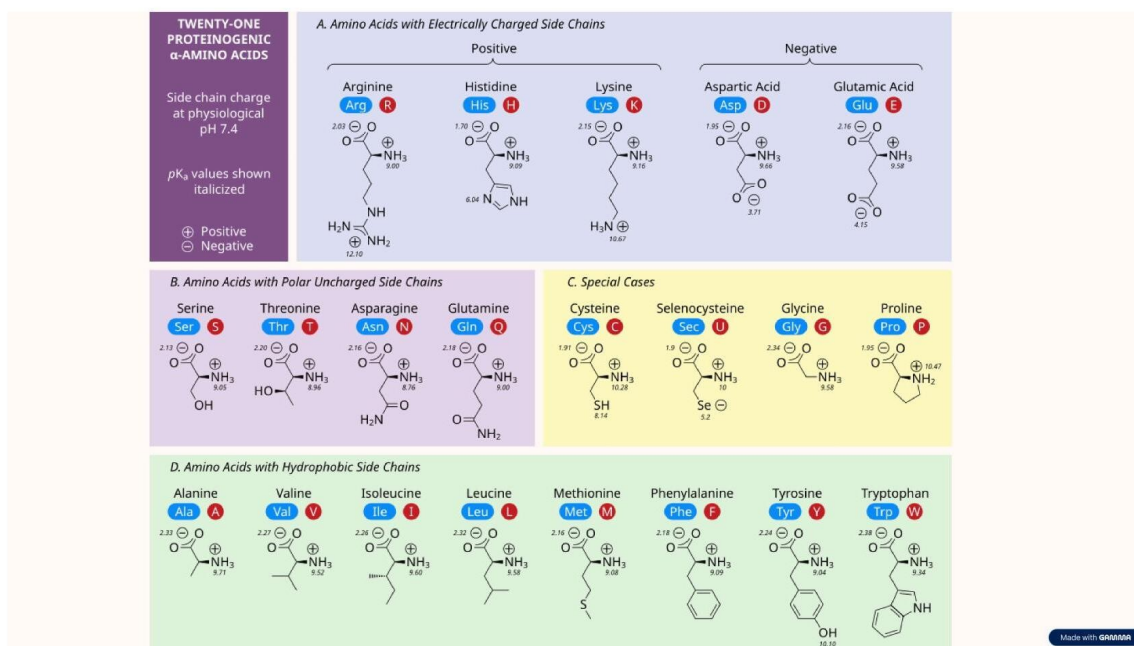


Fig. A19 — Structures of the 20/21 proteinogenic amino acids grouped by side-chain type, with three-letter and single-letter codes

A20. What is a peptide bond?

Ans: A peptide bond is a covalent amide bond formed between the α -carboxyl group ($-\text{COOH}$) of one amino acid and the α -amino group ($-\text{NH}_2$) of the next, via a dehydration-condensation reaction that releases one molecule of water. Successive peptide bonds link amino acids into a polypeptide chain.

A21. What do you mean by structural and functional proteins?

Ans: Structural proteins provide mechanical support and shape to cells and tissues, e.g., collagen (tendons, skin), keratin (hair, nails), and actin (cytoskeleton). Functional proteins perform dynamic biochemical roles such as catalysis, transport or signalling, e.g., enzymes (amylase), transport proteins (haemoglobin), and hormones (insulin).

A22. What kind of interactions are responsible for stabilisation of the secondary structure of a protein?

Ans: The secondary structure (α -helix and β -sheet) is stabilised by hydrogen bonds formed between the backbone $-\text{NH}$ and $-\text{C}=\text{O}$ groups of the polypeptide chain (i.e., between the amide hydrogen of one residue and the carbonyl oxygen of another), not by side-chain interactions.

Topic 7: Nucleic Acids (DNA & RNA)**A23. What is a nucleotide? Name its three components.**

Ans: A nucleotide is the monomeric building block of DNA and RNA. It consists of three covalently joined components: (i) a nitrogenous base (a purine — adenine/guanine — or a pyrimidine — cytosine/thymine/uracil), (ii) a pentose sugar (deoxyribose in DNA, ribose in RNA), and (iii) one or more phosphate groups.

A24. State Chargaff's rule of base pairing.

Ans: In any double-stranded DNA molecule, the amount of adenine equals the amount of thymine $[\text{A}] = [\text{T}]$, and the amount of guanine equals the amount of cytosine $[\text{G}] = [\text{C}]$. Consequently, the total purine content equals the total pyrimidine content.

Topic 8: Enzymes**A25. What are enzymes? Give two examples of enzymes found in the body.**

Ans: Enzymes are biological catalysts, predominantly proteinaceous in nature, that accelerate biochemical reactions by lowering the activation energy required, without themselves being consumed in the reaction. Examples: Amylase (breaks down starch into maltose) and Pepsin (breaks down proteins into smaller peptides).

Topic 9: Photosynthetic Pigments**A26. Define chlorophyll. Write down two functions of chlorophyll.**

Ans: Chlorophylls are the green pigments responsible for photosynthesis in plants, algae and cyanobacteria; each molecule contains a porphyrin ring with a central magnesium (Mg^{2+}) ion and a hydrocarbon phytol tail. Functions: (i) captures solar energy and converts it into chemical energy during photosynthesis; (ii) chlorophyll participates directly in the photochemical reactions of the light-dependent phase.

A27. Write down two functions of carotenoids.

Ans: (i) Act as accessory photosynthetic pigments, absorbing light wavelengths that chlorophyll cannot capture and passing the energy on. (ii) Protect chlorophyll from photo-oxidative damage by dissipating excess absorbed energy, contributing to the cell's antioxidant defence.

SECTION B — 7 MARKS QUESTIONS

(Answer in detail with diagrams wherever applicable, covering every topic of Unit 1)

B1. Describe the fundamental characteristics of living organisms.

Ans: Living organisms are distinguished from non-living matter by nine cardinal properties:

- Growth — an increase in cell number (hyperplasia) or cell size (hypertrophy); plants show indeterminate growth while animals generally stop growing at maturity.
- Reproduction — the ability to produce offspring, either asexually (one parent, identical offspring, e.g., binary fission in bacteria) or sexually (genetic recombination from two parents, e.g., meiosis in humans).
- Metabolism — the sum of all anabolic (build-up) and catabolic (break-down) biochemical reactions occurring continuously within the organism, with ATP as the energy currency.
- Cellular organisation — all life is composed of one or more cells; organisms may be prokaryotic (no nucleus) or eukaryotic (membrane-bound nucleus).
- Consciousness/Responsiveness — the ability to sense and respond to environmental stimuli, e.g., phototropism in plants, self-consciousness in humans.
- Homeostasis — the maintenance of a stable internal environment despite external fluctuations, e.g., the human body regulates temperature around 37°C via sweating and shivering.
- Adaptation — heritable structural, physiological or behavioural changes that improve survival, arising over generations through natural selection.
- Movement — external (locomotion) or internal (cytoplasmic streaming, vesicle transport).
- Life span — every organism has a definite period from birth/germination to death.

B2. Elaborate on the different types of organic constituents present in living cells, with examples.

Ans: Living cells are composed chiefly of three classes of organic macromolecules — carbohydrates, lipids and proteins — together with nucleic acids, each serving distinct structural and functional roles.

- Carbohydrates: polyhydroxy aldehydes/ketones with general formula $C_n(H_2O)_n$. Classified by degree of polymerisation into monosaccharides (glucose, fructose, galactose), disaccharides (sucrose, lactose, maltose), oligosaccharides (2–9 units, involved in cell recognition) and polysaccharides (starch, glycogen — energy storage; cellulose, chitin — structural support).
- Lipids: a structurally diverse group defined by solubility in non-polar solvents. Includes fatty acids, triacylglycerols (energy storage), phospholipids (membrane structure), steroids (e.g., cholesterol — hormone precursor), and lipoproteins (lipid transport in blood).
- Proteins: polymers of 20 standard amino acids linked by peptide bonds; serve as enzymes, structural scaffolds, transporters, receptors, antibodies and hormones. Structure is hierarchical — primary, secondary, tertiary and quaternary.
- Nucleic acids: DNA and RNA, polymers of nucleotides (base + pentose sugar + phosphate); store and express genetic information.

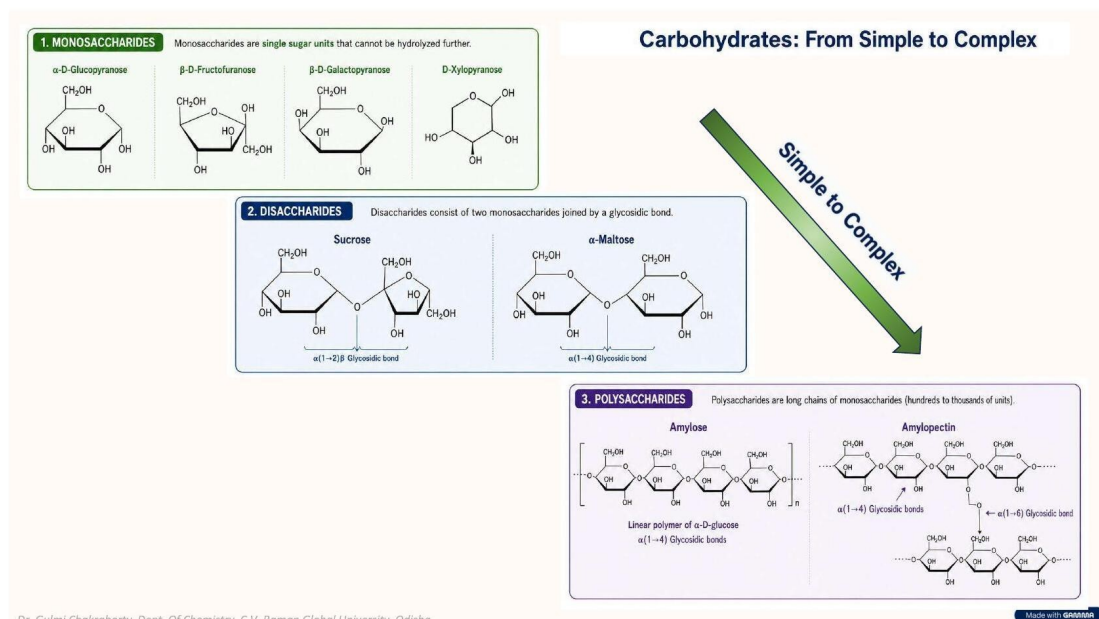


Fig. B2 — Carbohydrates: monosaccharides, disaccharides and polysaccharides, from simple to complex

B3. Compare and contrast prokaryotic and eukaryotic cells.

Ans: The prokaryote/eukaryote distinction is one of the most fundamental organising principles in biology:

Feature	Prokaryotic Cell	Eukaryotic Cell
Nucleus	Absent — DNA lies free in the nucleoid	Present — true, membrane-bound nucleus
Cell size	0.1 – 0.5 μm	10 – 100 μm (up to $\sim 200\times$ larger)
Organelles	Lack membrane-bound organelles	Possess mitochondria, ER, Golgi, etc.
Cellularity	Always unicellular	May be unicellular or multicellular
Ribosomes	Smaller, 70S	Larger, 80S
Reproduction	Mainly binary fission (asexual)	Sexual and asexual
Cell wall	Present in most	Present in plants/fungi; absent in animals
Examples	Bacteria, archaea, cyanobacteria	Plants, animals, fungi, protozoans

The transition from prokaryotic to eukaryotic cells is believed to have occurred about two billion years ago through endosymbiosis, where free-living prokaryotes were engulfed by and permanently integrated with a host cell.

B4. Describe the functions of the cell membrane, cell wall, and cytoplasm.

Ans: Cell (Plasma) Membrane:

- Support and Protection — separates the cell from its external environment and shields internal contents.
- Selective Permeability — structurally porous, it permits controlled movement of substances into and out of the cell.
- Boundary Formation and Communication — interfaces with adjacent cells and contains receptor proteins for chemical signalling; described by the Fluid Mosaic Model as a dynamic phospholipid bilayer with mobile embedded proteins.

Cell Wall:

- Composed of cellulose, hemicellulose and pectin; forms the rigid outermost layer in plant cells (also in fungi, some bacteria).

- Provides shape, structural support, and protection against mechanical and osmotic stress; freely permeable, unlike the plasma membrane.

Cytoplasm:

- The jelly-like substance (~80% water) between the plasma membrane and nucleus; site of most metabolic reactions including glycolysis and protein synthesis.
- Suspends organelles (ER, mitochondria, ribosomes, vacuoles) and contains the cytoskeleton (actin filaments, microtubules) for structural support and intracellular transport.

B5. Describe the structure and functions of cell organelles such as the endoplasmic reticulum, Golgi apparatus, mitochondria and lysosomes.

Endoplasmic Reticulum (ER):

- A network of membrane-bound sacs and tubules occurring in two forms. Rough ER (RER) is studded with ribosomes and is the principal site of protein synthesis, folding and quality control. Smooth ER (SER) lacks ribosomes and carries out lipid/steroid synthesis, detoxification of drugs/toxins, and calcium storage.

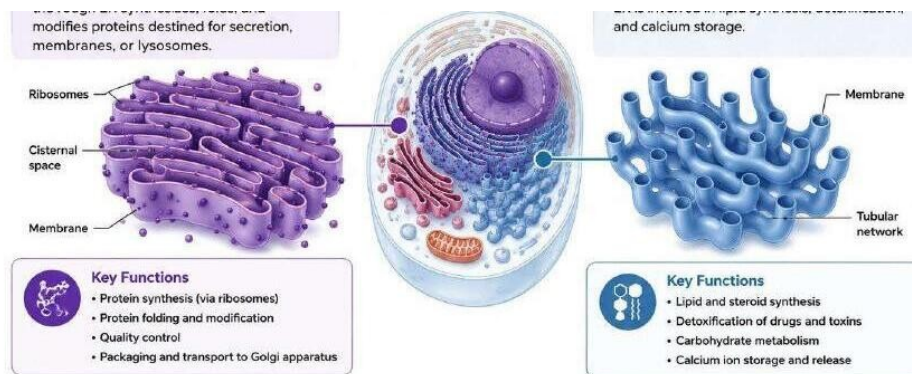


Figure 2.3: Structure of the Endoplasmic reticulum

The endoplasmic reticulum occurs in two structurally and functionally distinct forms. The Rough Endoplasmic

Fig. B5a — Rough ER vs. Smooth ER

Golgi Apparatus:

- A stack of membranous cisternae (cis face receives from ER, trans face ships out) that modifies (adds sugar chains), sorts, packages, and delivers proteins/lipids to the plasma membrane, lysosomes, or for secretion.

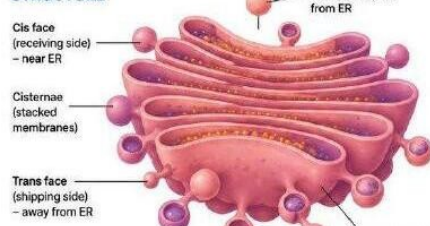
membrane. With the exception of sperm cells and red blood cells, the endoplasmic reticulum is present in essentially every type of eukaryotic cell.

2.6 The Golgi Apparatus

Golgi Bodies

The Golgi apparatus (Golgi bodies) is a membranous organelle that modifies, sorts, and packages proteins and lipids for transport to their final destinations inside or outside the cell.

STRUCTURE



FUNCTIONS

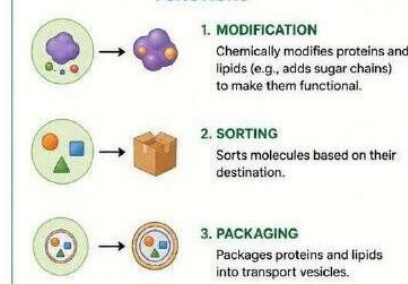


Fig. B5b — Structure and functions of the Golgi apparatus

Mitochondria:

- Double-membraned organelles that are the principal site of ATP generation via aerobic respiration — the "powerhouse of the cell"; contain their own DNA and ribosomes.

Lysosomes:

- Membrane-bound vesicles containing hydrolytic enzymes for intracellular digestion of waste and foreign material; termed "suicide bags" because rupture of the membrane can digest the cell itself (autolysis).

B6. What are the differences between plant cells and animal cells?

A recurring examination and conceptual theme in cell biology is the systematic comparison of plant, animal, and bacterial cells. The following consolidated tables present these comparisons across structural, biochemical, and physiological criteria.

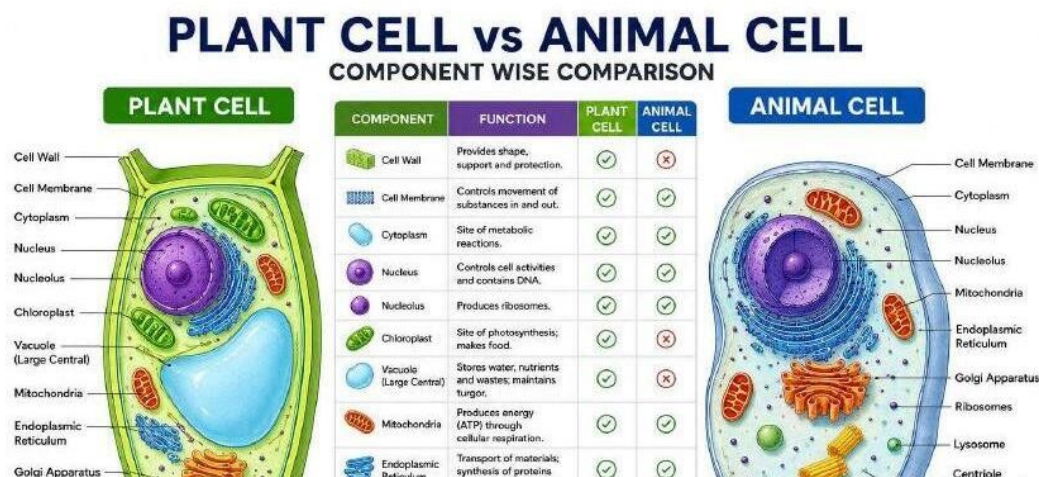


Fig. B6 — Comparative structure of plant and animal cells

Characteristic	Plant Cell	Animal Cell
Cell wall	Present (cellulose)	Absent
Shape	Fixed, rectangular	Flexible, round/oval
Chloroplast	Present	Absent
Vacuole	Large, single, central	Small or absent
Centriole	Absent	Present
Lysosomes	Few / rare	Present
Nutrition	Autotrophic	Heterotrophic

B7. Write down the differences between the animal cells and bacterial cells.

Characteristic	Animal Cell	Bacterial Cell
Cell type	Eukaryotic	Prokaryotic
Nucleus	Present (true nucleus)	Absent (nucleoid instead)
Cell size	10 – 100 μm	0.2 – 2 μm
Cell wall	Absent	Present
Ribosomes	Larger, 80S	Smaller, 70S
Mitochondria	Present	Absent
Plasmids	Absent	Present
Mode of respiration	Aerobic	Aerobic and anaerobic

B8. Define cell. Write down five characteristics of cells. Draw a structure of the cell with a labelled diagram.

Ans: A cell is the smallest structural and functional unit of life, capable of independent existence and carrying out all essential life processes.

Five characteristics of cells:

- Cells are the lowest level of structural organisation common to every living system, and house the organism's hereditary material (DNA).
- Cells range enormously in size — from $\sim 0.1 \mu\text{m}$ (Mycoplasma) to $\sim 150 \mu\text{m}$ (ostrich egg) — yet share a common organisational plan.
- Cells may be prokaryotic (no true nucleus) or eukaryotic (membrane-bound nucleus).
- Cells carry out metabolism, respond to stimuli, grow, and reproduce.
- Cells contain specialised organelles, each performing a distinct function essential to the cell's survival.

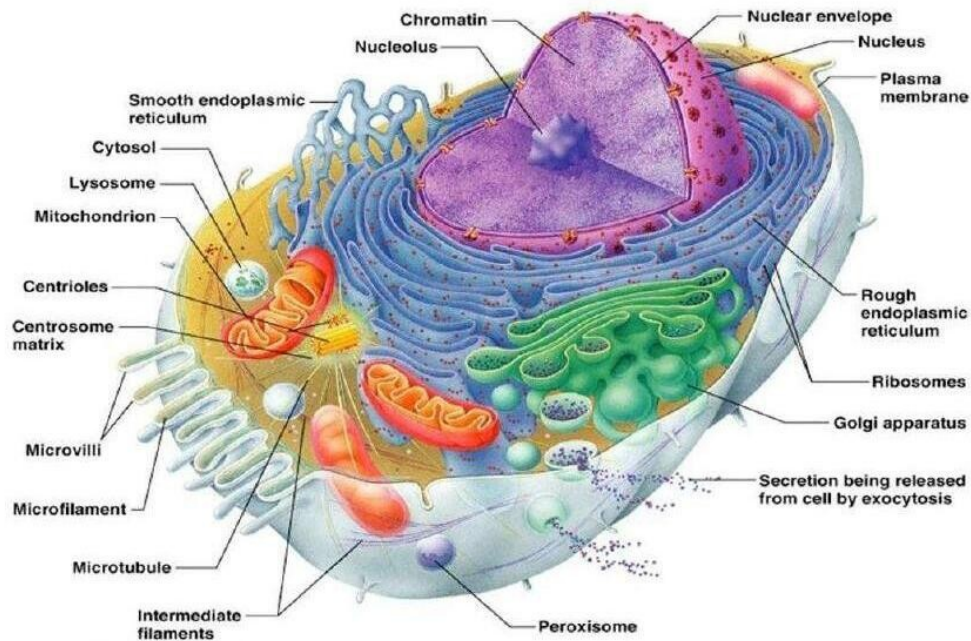


Figure 2.1: Generalised eukaryotic cell showing major organelles.

Fig. B8 — Generalised eukaryotic cell showing major organelles

B9. What are enzymes? Discuss their structure, characteristics and working mechanism in the body.

Ans: Enzymes are biocatalysts, predominantly proteinaceous, that accelerate biochemical reactions within living systems by lowering the activation energy, without being consumed in the process.

Structure:

- A complete functional enzyme (holoenzyme) comprises an apoenzyme (the protein component, catalytically inactive alone) and, in many cases, a cofactor (a non-protein component — a metal ion or organic molecule).
- A specific region called the active site is where the substrate binds and catalysis occurs.

Characteristics:

- Highly specific for their substrate (Lock-and-Key / Induced Fit model); accelerate reactions without being consumed; sensitive to temperature and pH (e.g., optimal $\sim 37^\circ\text{C}$ for human enzymes; pepsin works best at pH 2); may require cofactors/coenzymes (e.g., Zn, Mg, NAD⁺); susceptible to competitive and non-competitive inhibition.

Mechanism of action:

- Step 1 — Substrate binding: the enzyme binds a specific substrate at the active site, forming an enzyme–substrate complex.
- Step 2 — Catalysis: the enzyme lowers the activation energy and facilitates the chemical transformation of the substrate.
- Step 3 — Product formation: product(s) are released and the enzyme is regenerated unchanged, ready for further reaction cycles.

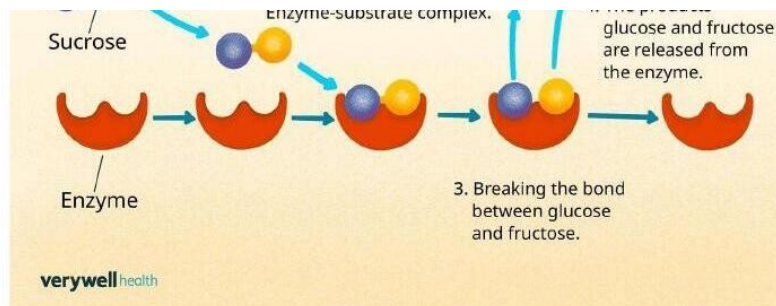


Figure 5.2: Enzyme mechanism — substrate binding, catalysis, and product formation.

Step 1 — Substrate Binding: At the active site, the enzyme binds to a specific substrate molecule, forming an enzyme-substrate complex.

Step 2 — Catalysis: The enzyme facilitates the chemical transformation, either breaking down the substrate (catabolism) or synthesising a new product (anabolism).

Fig. B9 — Enzyme mechanism: substrate binding, catalysis and product formation (example: sucrose → glucose + fructose)

Physiological examples: Amylase (starch → maltose), Pepsin (proteins → peptides), Lipase (fats → glycerol + fatty acids).

B10. Distinguish between primary, secondary, tertiary and quaternary structures of protein.

Ans: Protein structure is hierarchical — each level builds upon the previous, and disruption of any level (denaturation) typically abolishes biological function.

Level	Description
Primary	The linear sequence of amino acids joined by covalent peptide bonds, read N-terminus to C-terminus; encoded directly by the gene.
Secondary	Local, regular folding patterns stabilised by hydrogen bonds between backbone -NH and -C=O groups: the α -helix (right-handed coil) and β -sheet (extended, laterally H-bonded strands).
Tertiary	The overall 3-D folding of the entire polypeptide chain, stabilised by hydrophobic interactions, hydrogen bonds, ionic bonds, disulphide bridges (-S-S-) and van der Waals forces; creates the active site.
Quaternary	The association of two or more polypeptide subunits into a single functional complex, e.g., haemoglobin (2 α + 2 β chains), stabilised by the same non-covalent forces plus inter-subunit disulphide bonds.

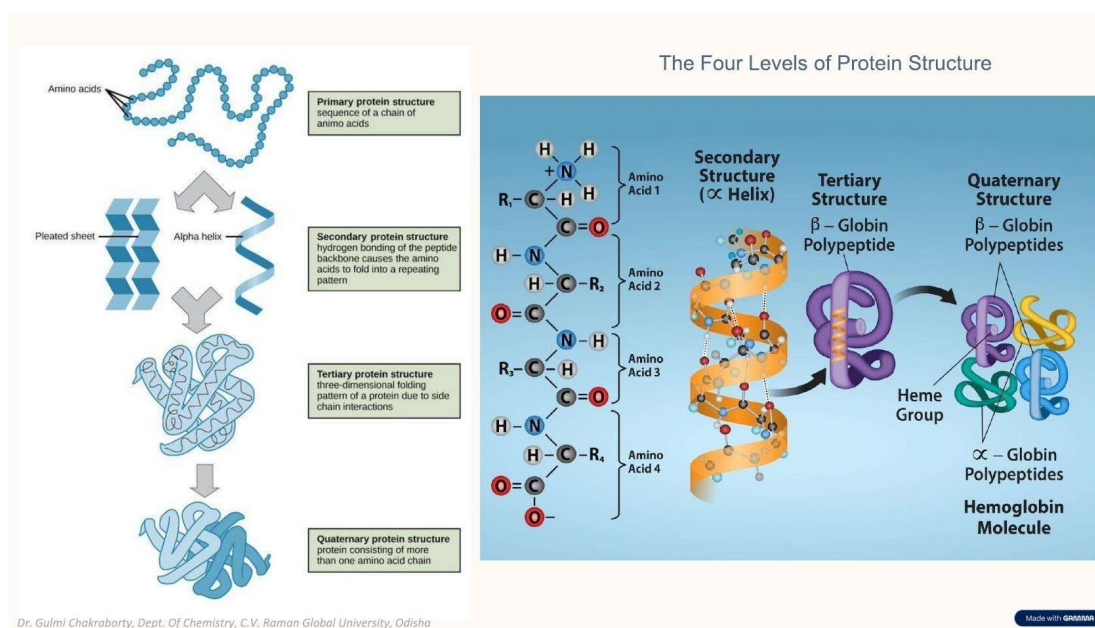


Fig. B10 — The four levels of protein structure, illustrated with haemoglobin

B11. Depict the primary structure of the following peptide codes: FYP, ATF, DCP.

Ans: The primary structure of a peptide is its linear sequence of amino acids, read from the free N-terminus (amino end) to the free C-terminus (carboxyl end), with successive residues joined by peptide bonds (formed by dehydration condensation between the -COOH of one residue and the -NH₂ of the next).

- FYP = Phe – Tyr – Pro → H₂N–Phenylalanine–Tyrosine–Proline–COOH (tripeptide, 2 peptide bonds)
- ATF = Ala – Thr – Phe → H₂N–Alanine–Threonine–Phenylalanine–COOH (tripeptide, 2 peptide bonds)
- DCP = Asp – Cys – Pro → H₂N–Aspartic acid–Cysteine–Proline–COOH (tripeptide, 2 peptide bonds)

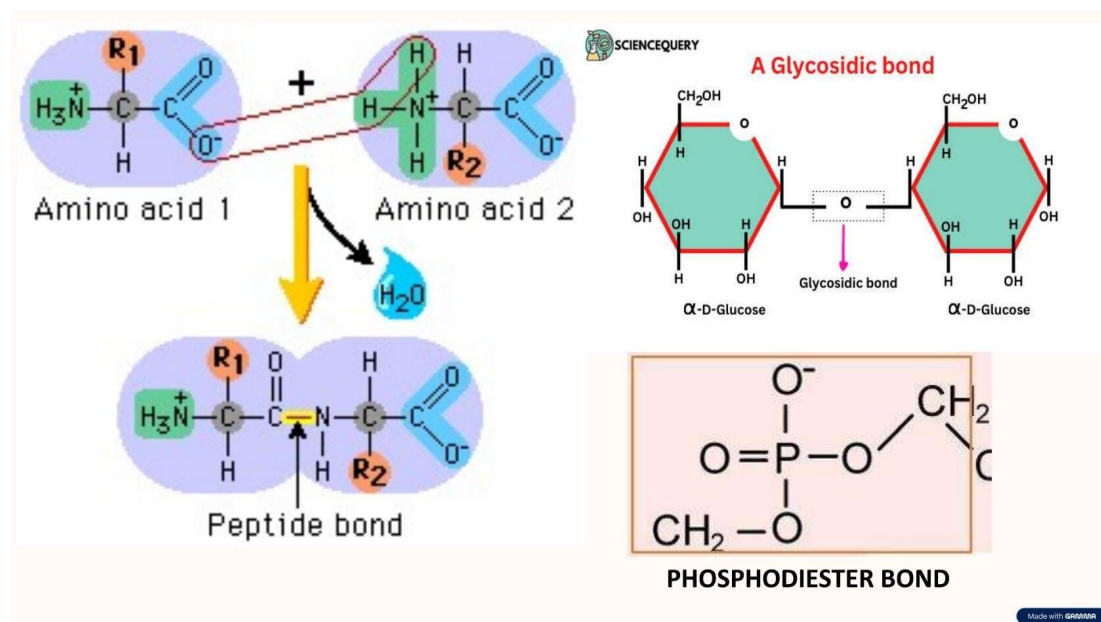


Fig. B11 — Peptide bond formation between two amino acids (dehydration condensation, releasing H₂O)

B12. Classify animal tissues with their functions.

Ans: A tissue is a group of cells with similar shape and function. Animal bodies are composed of four fundamental tissue types:

Tissue Type	Function / Example
Epithelial tissue	Covers body surfaces and lines internal cavities and organs, e.g., skin, gut lining.
Connective tissue	Supports and binds other tissues together, e.g., bone, blood, cartilage.
Muscular tissue	Responsible for movement; comprises skeletal, smooth and cardiac muscle.
Nervous tissue	Transmits electrical (nerve) signals, e.g., brain, spinal cord, peripheral nerves.

These tissues combine to form organs (e.g., heart, liver), which in turn form organ systems, giving the hierarchical organisation: Cells → Tissues → Organs → Organ Systems → Organism.

B13. Describe the structure of the DNA double helix (Watson–Crick model) and differentiate DNA from RNA.

Ans: James Watson and Francis Crick proposed the double helix model of DNA in 1953, using X-ray diffraction data obtained by Rosalind Franklin and Maurice Wilkins.

Key structural features:

- A nucleotide (the monomer) consists of a nitrogenous base (purine — adenine/guanine; or pyrimidine — cytosine/thymine), a pentose sugar (deoxyribose), and a phosphate group.
- Two antiparallel polynucleotide strands wind into a right-handed helix; the sugar-phosphate backbone (linked by 3'-5' phosphodiester bonds) lies on the outside, with nitrogenous bases pointing inward.

- Complementary base pairing (Chargaff's rule): Adenine pairs with Thymine via 2 hydrogen bonds; Guanine pairs with Cytosine via 3 hydrogen bonds.
- The helix has a pitch of 34 Å per turn (10 base pairs/turn) and displays a major groove and a minor groove, which are binding sites for regulatory proteins.

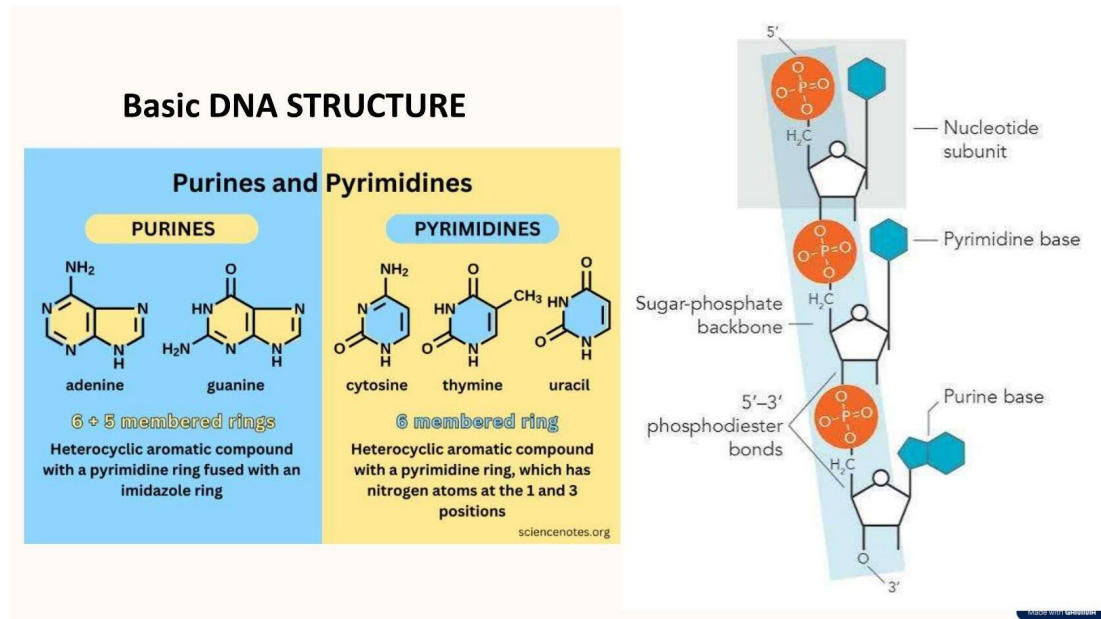


Fig. B13a — Purine and pyrimidine bases; the nucleotide sugar-phosphate backbone

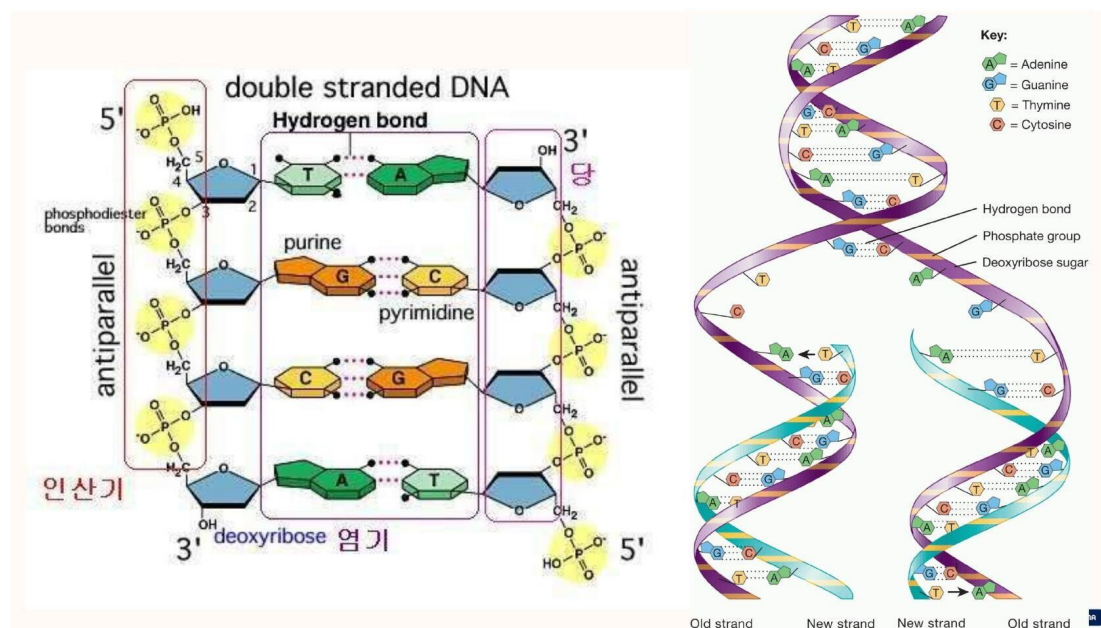


Fig. B13b — The DNA double helix showing antiparallel strands, base pairing and replication

DNA vs. RNA:

Feature	DNA	RNA
Sugar	Deoxyribose (no 2'-OH)	Ribose (has 2'-OH)
Bases	A, T, G, C	A, U, G, C (U replaces T)
Strands	Usually double-stranded	Usually single-stranded
Stability	Very stable (no 2'-OH)	Less stable (2'-OH promotes hydrolysis)
Function	Genetic information storage	Gene expression (mRNA, tRNA, rRNA)

B14. What are chromosomes and genes? Explain the organisation of DNA into chromatin and chromosomes.

Ans: A gene is a specific segment of DNA that carries the coded instructions for the synthesis of a functional product — usually a protein or an RNA molecule — and represents the basic unit of heredity. A chromosome is a highly condensed, organised structure of DNA tightly wound around histone proteins, in which the genetic material (genes) of an organism is packaged; it becomes visible as a distinct rod-like structure during cell division.

Organisation of DNA into chromatin and chromosomes:

- Within the nucleus, DNA associates with basic histone proteins to form chromatin — a bead-on-a-string arrangement of DNA wrapped around histone octamers (nucleosomes).
- During interphase, chromatin remains loosely dispersed (allowing gene transcription); prior to cell division, chromatin coils and condenses further into visible, discrete chromosomes to ensure accurate segregation of genetic material to daughter cells.
- The human genome comprises approximately 3 billion base pairs of DNA, organised into 46 chromosomes (23 pairs) and encoding roughly 20,000–25,000 genes.
- The nuclear envelope (a double membrane with nuclear pores) separates and protects this DNA from the rest of the cell while permitting controlled exchange of molecules.